MLOPS PROJECT PROPOSAL · SPRING 2026

RentIQ

Intelligent apartment recommendations powered by multi-signal ML fusion — pricing fairness, NLP landlord scoring, and neighborhood profiling.

MLOPS | NLP | SF BAY AREA

Gyoungwon Jeong · Mani Atwal · Shiman Zhang

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THE PROBLEM

Apartment search is broken for renters.

Static filters. No pricing fairness. No landlord quality data. No personalization.

RentIQ is an end-to-end ML pipeline that turns raw rental data into personalized apartment recommendations — combining pricing fairness, landlord quality scores, and neighborhood signals.

SCOPE

San Francisco Bay Area · ~1,000 listings.

03 What RentIQ Does

01	Structured Listing Extraction	Price, bedrooms, amenities, availability
02	Pricing Analysis	Predict fair market rent · flag overpriced listings
03	Landlord Quality Scoring	NLP sentiment analysis on property reviews
04	Neighborhood Profiling	Walkability, transit, amenities, crime context
05	Personalized Ranking	ML recommendations matched to user preferences

04 Competitive Landscape

PLATFORM	KEY STRENGTH	CRITICAL GAP	OUR ADVANTAGE
Zillow / Trulia	Massive inventory, Zestimate	No landlord quality signal	+ Landlord scoring + overpricing detection
Apartments.com	Good listing density, reviews	Reviews unused as ML signal	+ ML-driven personalization from reviews
Redfin	Price history, transparent fees	Primarily for-sale focus	+ Renter-first + landlord quality layer
ApartmentList	Quiz-based matching, renter UX	Black-box, no overpricing signal	+ Transparent, explainable scores

Only RentIQ combines: **pricing fairness + landlord quality + ML-personalized recommendations**

05 SWOT Analysis

STRENGTHS

- Multi-source data fusion — pricing + reviews + neighborhood
- Landlord quality scoring (underserved in all existing tools)
- Personalized ML recs vs. static search filters
- Overpricing detection — unique, actionable insight

WEAKNESSES

- Apartments.com scraping brittle — anti-bot (Akamai WAF)
- Crime data city-level only; no neighborhood granularity
- Sparse reviews for newer / smaller-market properties
- Landlord entity resolution hard across name variations

OPPORTUNITIES

- Renters increasingly search and decide online
- No existing tool combines pricing + landlord quality
- LLMs enable powerful NLP on unstructured reviews
- WalkScore & Redfin APIs free and well-documented

THREATS

- ToS changes can disable scraping at any time
- Zillow / Redfin have massive data moats
- Rental market moves fast — static scrapes go stale
- Historical pricing bias can propagate through ML models

06 Datasets & Acquisition

LISTINGS & PRICING

Zillow ZORI / ZHVI
Direct CSV download — regional pricing benchmarks

Apartments.com (scraper)
~1,000 SF listings — price, amenities, availability

US Census ACS API
Median rent, income, vacancy rates by zip/tract

LANDLORD REVIEWS

Google Places API
$200/mo free credit — ratings & review excerpts

ApartmentRatings.com
Custom scraper — renter-specific landlord feedback

NEIGHBORHOOD SIGNALS

WalkScore API
Free · 5,000 calls/day — Walk / Transit / Bike scores

OpenStreetMap (Overpass)
Fully free — parks, transit stops, grocery, cafes

FBI Crime API (UCR)
Free (data.gov key) — city-level crime statistics

07 1-Month Roadmap

WEEK 1
Data Collection & EDA

- → Download Zillow ZORI/ZHVI + Redfin CSVs
- → Pull Census ACS data via API
- → Scrape 500–1,000 SF Apartments.com listings
- → Enrich with WalkScore + OSM amenities

500+ listings · no missing geo fields

WEEK 2
Feature Engineering & NLP

- → Scrape ApartmentRatings + Google Places reviews
- → Run sentiment analysis (HuggingFace / OpenAI)
- → Build landlord quality scores
- → Engineer $/sqft, ZORI delta, price features

Sentiment labels for 80%+ of listings

WEEK 3
ML Modeling

- → Train rent predictor (XGBoost / Random Forest)
- → Build overpricing detector
- → Build recommendation engine
- → Evaluate: MAE on rent, precision@k on recs

Rent MAE <15% · Precision@5 >0.6

WEEK 4
Evaluation & Demo

- → Build lightweight demo UI
- → Run ablation study on features
- → Document pipeline + model cards
- → Final write-up and slide deck

End-to-end demo: prefs → ranked apartments

Let's Build RentIQ

An end-to-end ML pipeline that gives renters what they've never had — pricing fairness, landlord transparency, and personalized recommendations.

DATA PIPELINES | NLP | XGBOOST | REC SYSTEMS | SF BAY AREA | MLOPS

Gyoungwon Jeong
Mani Atwal
Shiman Zhang